TALHA GÜRLER

Kocaeli University • Computer Eng.

Computer Engineering Student • AI/ML, Quantitative FinTech & Full-Stack Software Developer

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PROFESSIONAL SUMMARY

Results-driven Computer Engineering student at Kocaeli University with extensive hands-on experience architecting production-grade systems across **Artificial Intelligence & Machine Learning (Multimodal RAG, Computer Vision, Time-Series & XGBoost Classification)**, **Quantitative Finance & Algorithmic Trading (Binance Futures Crypto & Global Equities)**, and **Full-Stack / Mobile Engineering (Python, TypeScript, React, Java, C)**. Proven track record of building autonomous crypto futures execution engines with order-book microstructure features, single-stock deep analysis & cross-sectional factor models validated on QuantConnect, offline-first PWA/Android applications, and low-level systems simulators.

TECHNICAL SKILLS & TOOLING

Programming Languages: Python, TypeScript, JavaScript (ES6+), C (C99), Java (Java 11+), SQL (SQLite, PostgreSQL), HTML5, CSS3

AI, Machine Learning & Data Science: XGBoost, Random Forest, Scikit-Learn, Optuna (Hyperparameter Optimization), Triple-Barrier Labeling, Brier Skill Calibration, LLM & Multimodal RAG, LangChain (LCEL), ChromaDB (Vector DB), Google Gemini API (2.5/2.0 Flash Vision & Embeddings), OpenAI Whisper (ASR), Hugging Face BERT, TensorFlow/Keras (Bi-LSTM), From-Scratch KNN & CART Decision Trees (Gini Impurity)

Algorithmic Trading & Quantitative Finance: Binance Futures API (CCXT), Order Book Imbalance (`OB_Imbalance`), Funding Rate & Open Interest Analytics, Interactive Brokers (TWS), QuantConnect (LEAN), SEC EDGAR Point-in-Time Fundamentals Pipeline, Walk-Forward Backtesting, Pandas, NumPy, SciPy, Apache POI, 3NF Database Normalization

Web, Mobile, Systems & Security: React 19, Vite 8, Tailwind CSS 4, Zustand, Dexie.js (IndexedDB), Capacitor 8 (Android APK), Flask (SSE Streaming), Streamlit, Java Swing (SwingWorker, Graphics2D), PyQt5, Selenium, Paramiko SSH, NIST PBKDF2-SHA256, RFC 6238 TOTP 2FA, Application-Layer WAF, Git, GitHub Actions, Vitest

PROJECTS & ENGINEERING EXPERIENCE

StockPredictionPro & QuantBot — Dual-Engine (Crypto Futures & Equities) AI Trading, Single-Stock Analysis & FinTech Platform

github.com/talha4111/StockPredictionPro ↗

Python | XGBoost | Optuna | Binance Futures (CCXT) | Order Book Imbalance | Streamlit (10-Page) | QuantConnect | SEC EDGAR PIT | TOTP 2FA & WAF

- **Binance Futures XGBoost & Execution Engine (QuantBot):** Engineered per-symbol **XGBoost** directional classifiers (`n_estimators=500`, Walk-Forward & Brier Skill gated) integrated with **Binance Futures (CCXT)** and **Interactive Brokers (TWS)** using **Order Book Imbalance (`OB_Imbalance`)**, **Funding Rate**, **Open Interest**, `CHOP`, `ADX`, and `Volume_ZScore` features, paired with dynamic drawdown margin scaling (`1.0x → 0.30x`), 24h **Symbol Guard** quarantine, single Hard-Stop synchronization, Binance Income Ledger (`REALIZED_PNL`, `COMMISSION`, `FUNDING_FEE`) reconciliation, and **Optuna** hyperparameter tuning.
- **Single-Stock Deep Analysis, Investment Committee & Dip Hunter:** Built a comprehensive single-stock decision engine featuring 4-pillar factor radar charts, an automated **Factor Conflict Detector** (flagging value traps vs. overextended momentum), a **Catalyst & Earnings Lab**, ATR-based staged entry/SL/TP calculators, and a **Triple-Barrier (±1.8 ATR) Hybrid AI** Dip & Profit-Protection Hunter calibrated on **2,837 out-of-sample events** with instant Telegram alerts.
- **4-Pillar Cross-Sectional Factor Engine (v3.1) & 10-Page Secured Web Platform:** Ranked 100+ US equities across *Quality (35%)*, *Value (20%)*, *Momentum (30%)*, and *Risk (15%)* using **Point-in-Time SEC EDGAR** filings and **QuantConnect** walk-forward validation (2021–2026); deployed a 10-page Streamlit platform with FIFO lot accounting, S&P 500 Beta/crisis stress testing, capital gains tax indexing, **NIST PBKDF2-SHA256**, **RFC 6238 TOTP 2FA**, and a sliding-window **Application-Layer WAF**.

Makro Takip (sporproje) — AI Vision Athlete Nutrition, Macro Gap Solver & Offline-First PWA / Android App

github.com/talha4111/sporproje ↗

React 19 | TypeScript 6 | Tailwind CSS 4 | Gemini 2.5 Flash Vision | Dexie.js (IndexedDB) | Zustand | Capacitor 8 APK | Vitest

- **AI Vision Plate Analyzer & Macro Gap Solver:** Built a client-side image compression and **Gemini 2.5 Flash Vision** pipeline estimating portion gram weights (`estimatedGrams`) and macro/micronutrients from meal photos, paired with a deterministic mathematical solver (`solveMacroGap`) that scales athlete meal templates to the exact gram.
- **Barcode Health Radar & Offline-First Mobile Architecture:** Integrated `html5-qrcode` and **Open Food Facts API** with an automated Health Radar auditing NOVA processing groups (1–4), Nutri-Score (A–E), and E-code additives; persisted data via `Dexie.js` and packaged as an **Android APK via Capacitor 8**.

EduBrain-AI — Multimodal Vision & ChromaDB Vector Database RAG Educational Assistant

github.com/talha4111/EduBrain-AI ↗

Python | LangChain (LCEL) | ChromaDB | Gemini 2.0 Vision | Streamlit | Poppler PDF2Image

- **Multimodal Extraction & Hybrid Chunking RAG:** Designed a document ingestion pipeline converting PDFs and handwritten lecture note images into structured Markdown via **Gemini 2.0 Flash Vision**, splitting via `MarkdownHeaderTextSplitter` and `RecursiveCharacterTextSplitter` into **ChromaDB** (`gemini-embedding-001`) for multi-turn conversational Q&A with source citations.

LingoReader AI (LingoProje) — Real-Time SSE Streaming AI Tutor & Whisper Karaoke Platform

github.com/talha4111/LingoProje ↗

Python | Flask SSE Streaming | OpenAI Whisper ASR | Gemini 2.5 Flash | yt-dlp & FFmpeg | JavaScript SPA

- **Timestamped YouTube ASR & SSE Streaming Tutor:** Engineered an audio transcription pipeline using `yt-dlp` and **OpenAI Whisper** for synchronized karaoke lyrics, paired with a Flask **Server-Sent Events (`text/event-stream`)** tutor injecting active screen context and 12-turn memory across multi-model Gemini fallbacks.

Multi-Technician Task Scheduling Simulator (PROLAB1) — Systems Programming & Heuristic Optimization

github.com/talha4111/PROLAB1 ↗

C (C99) | GCC | OS Scheduling (SJF, RR, Priority) | Heuristic Optimization | IEEE Paper

- **Custom Location-Aware Heuristic (-15.0% Makespan):** Architected a modular C99 simulator modeling 4 parallel technicians across 5 rooms with a 1-unit transition penalty and deterministic time-travel state replay (`zamana_git`); designed a custom heuristic (`gorev_sec_ozgun`) that reduced 100-task benchmark makespan from **200 to 170 units (-15.0%)** with **96.76% resource utilization**.

Sales Predictor (prolab2), JSON-to-SQL 3NF Converter (prolab3) & Autonomous YHT Ticket Bot

prolab2 • prolab3 • yht-ticket-bot ↗

Java 11+ & Swing | From-Scratch KNN & CART | Apache POI | PyQt5 & SQLite 3NF | Selenium & Paramiko SSH

- **From-Scratch Java ML & Autonomous 3NF Normalizer:** Implemented pure-Java `PriorityQueue` Euclidean **KNN** and recursive **Gini Impurity CART Decision Tree** classifiers with a `SwingWorker` GUI (`prolab2`); a 9-module Python & PyQt5 tool converting nested schema-less JSON into **3NF relational SQLite tables** (`prolab3`); and a 24/7 headless Selenium ticket scanner with Telegram alerts and `Paramiko` Linux VPS deployment.

EDUCATION

Kocaeli University — Faculty of Engineering, B.Sc. in Computer Engineering

Kocaeli, Türkiye
2023 – Present

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Programming Laboratory I & II, Database Management Systems, Discrete Mathematics.